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**Повышение справедливости в системах кадрового отбора на
основе ИИ: интерпретируемая система скрининга резюме**

Topic

**Improving Fairness in AI-Powered Recruitment: An Interpretable
Resume Screening System**

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Chapter 1

Introduction

1.1 Relevance of the Topic

The use of AI and ML in recruitment has moved from experimental pilots to actual HR pipelines in the last few years. Companies now rely on automated systems to screen large volumes of CVs, to match candidates to vacancies, and to support recruiters with rankings and shortlists. Early work like PJFNN by Zhu et al. [1] (2018) already showed that deep models can learn good resume – job representations, so the technical foundation has been there for a while.

What has changed is the scale and the expectations. AI tools are now often presented as a way to hire faster, to reduce human subjectivity, and to make candidate evaluation “more objective.” However, we already know from other large platforms that algorithmic systems can produce skewed outcomes even when no explicit bias is coded in. Ali et al. [2], for example, showed on Facebook’s ad delivery that gender and racial skews still appeared under neutral targeting.

This is especially relevant for hiring, because recruitment data is deeply historical: it reflects labour market inequalities, company preferences, and social

patterns. Bias can appear at several levels: among people, in the data, in the learning process, and in deployment [3].

The central problem is that AI systems in hiring are introduced to make decisions fairer and more consistent, but in practice they can reproduce or even amplify existing disparities. Models learn from past decisions, and past decisions are not neutral.

Historical HR datasets may reflect preferences for certain universities, cities, genders, or career paths. Even when protected attributes are removed, related features can still act as proxies. This is why bias is often described as a systematic tendency to favour one group over another, directly or indirectly [3].

Explainable AI (XAI) is often proposed as a way to make this visible. Methods such as SHAP, LIME, and Integrated Gradients can show which features actually drive the prediction, and this helps detect suspicious dependencies. However, explainability does not guarantee fairness.

Fairness is multi-dimensional. Metrics like demographic parity, equalized odds, and counterfactual fairness capture different normative goals, and they cannot always be optimized at the same time [4]. Mehrabi et al. [5] discuss how XAI can help connect model behaviour to fairness analysis, while Geyik et al. [6] show that, even after bias is detected, we may still need to re-rank candidates to obtain a fairer output list.

Similar issues have been observed in other high-stakes domains. Chouldechova [7] demonstrated that criminal justice models can be accurate and still unfair across groups. Raghavan et al. [8] traced how bias can enter very early through pre-employment assessments. Recruitment, therefore, faces the same pattern: we have detection tools and we have mitigation tools (pre-processing, in-processing, post-processing [9], [10]), but we do not yet have one stable, deployment-ready

workflow that HR teams can use while also complying with GDPR and anti-discrimination rules.

This matters because hiring decisions are consequential. If an AI system decides which CVs a human will see, then any bias at that stage becomes a gate-keeping mechanism. Organizations must be able to explain why a model preferred one candidate over another and must show that protected groups were not systematically disadvantaged.

At the same time, organizations do not want to lose the advantages of automation: speed, consistency, and sometimes better matching. Existing mitigation strategies (rebalancing data before training, adding fairness constraints to the model, or adjusting rankings afterwards) do improve fairness, but they all introduce trade-offs between fairness and performance [5], [6].

The present thesis applies an *audit-first* procedure on real Russian IT resumes: measure group disparities, use Integrated Gradients to locate salient tokens, benchmark debiasing families, and report utility, TPR-gap fairness, and city-swap wording robustness as separate axes (Sections 1.2–1.6). The empirical focus is geographic proxy bias in text, while gender and age are kept in the analysis for comparison.

1.2 Purpose of the Study

The purpose of this study is to design, implement, and evaluate a repeatable *five-phase audit protocol* for BERT-based multi-class IT resume screening on Russian HeadHunter data [11]. The protocol moves from baseline disparity measurement and token-level diagnosis to debiasing benchmarks, selected combinations, validation runs, and a consolidated trade-off summary (Section 3.8).

Scope of sensitive attributes. Resumes carry city, gender, and age metadata for evaluation only. Phase 1 compares robust true positive rate (TPR) gaps across all three attributes; **geographic disparity is the dominant signal** on the baseline (Chapter 4). Later debiasing experiments therefore target city-related bias (41 city groups, city-swap tests, city-targeted training interventions), while gender and age gaps stay in the report as comparative context.

Results are reported on three evaluation axes (Section 3.4): utility (accuracy and macro F1), group fairness (robust worst-case and macro-average TPR gaps), and wording robustness under the city-swap protocol (flip rate on a named subset of checkpoints; Section 3.5.4). The goal is to spell out the accuracy, fairness, and robustness trade-offs openly. We do not argue that one model wins on every criterion at once.

1.3 Object and Subject of Research

The **object of research** is automated resume screening based on deep text classifiers trained on real HR data.

The **subject of research** is methods for *measuring and mitigating geographic proxy bias* in such systems, together with comparative measurement of gender- and age-related TPR gaps on the same corpus. This includes explainable AI for pre-debiasing diagnosis (Integrated Gradients), group fairness metrics, geographic counterfactual evaluation (city-swap), and debiasing techniques applied during training or, in one setup, post hoc on classifier scores.

1.4 Research Tasks

The following tasks support the stated purpose:

1. Review prior work on bias, fairness, and explainability in AI-supported recruitment (Chapter 2).
2. Construct a nine-class resume dataset and a BERT baseline with inverse square-root reweighting by city group (Chapter 3).
3. Run Phase 1 of the audit protocol: robust TPR gaps by city, gender, and age on the baseline; Integrated Gradients (IG) attributions to build scrubbing and city-swap lexicons.
4. Run Phases 2–3: implement, tune, and compare debiasing families and selected combinations, with city-swap on strong checkpoints.
5. Run Phases 4–5: check reproducibility under additional random seeds and post-hoc logit adjustment; summarise axis-specific optima and limitations (Chapters 4–5).

1.5 Scientific Novelty

The main novel elements are:

1. A documented **five-phase audit protocol** (diagnostic measurement; debiasing benchmark; combinations and geographic counterfactuals; validation; trade-off consolidation) on 27,550 Russian IT resumes across 41 city groups [11].

2. An empirical comparison of more than 25 debiasing configurations that links IG token attributions to group-level TPR gaps and, for selected models, city-swap flip rates.
3. Explicit **three-axis** trade-offs (utility, group fairness, geographic wording robustness), showing that leading configurations differ by deployment priority.
4. Documented limits of standard techniques in a high-cardinality geographic setting: weak adversarial debiasing across many city groups; class-rebalancing losses with little effect on geographic gaps; post-hoc logit adjustment that can raise F1 while widening fairness gaps.

1.6 Research Questions

The study addresses three research questions:

RQ1: On Russian IT resume data, does a BERT-based classifier show measurable TPR disparities across city groups, and how do these compare with gender- and age-based gaps on the same baseline?

RQ2: Do IG attributions for city- and age-related tokens line up with group-level TPR gaps from Phase 1 as diagnostic evidence (consistency checks only)?

RQ3: How do debiasing configurations differ on utility, group fairness, and city-swap flip rate, and which choice fits which deployment priority?

1.7 Working Propositions

We set out four testable propositions for the experiments:

1. **P1 (Bias inheritance).** The baseline will show group-level TPR disparities; geographic gaps are expected to exceed gender- and age-based gaps on the same model.
2. **P2 (Diagnostic linkage).** IG attributions for city and age tokens will align directionally with group-level gaps and inform scrubbing and city-swap design.
3. **P3 (Conditional mitigation).** Debiasing families will improve some evaluation axes while leaving others unchanged or worse.
4. **P4 (No universal optimum).** No tested configuration will rank first simultaneously on macro F1, worst-case TPR gap, and city-swap flip rate.

Chapter 2

Literature Review

This chapter reviews explainable AI, fairness definitions, and debiasing methods in recruitment. It closes with propositions P1–P4 and the three-axis evaluation setup used in the experiments.

2.1 Explainable AI in Recruitment

The concept of Explainable AI (XAI) has gained significant traction in recent years, especially in fields like healthcare, finance, and recruitment, where automated decisions have far-reaching impacts [12], [13]. XAI includes methods designed to provide clear explanations for the predictions made by machine learning models, which is crucial in recruitment, where AI decisions affect careers. Chen [14] argues that AI can enhance recruitment quality and efficiency, but risks perpetuating bias based on gender, race, geography, or other traits.

A major challenge in recruitment AI is the increasing complexity of the models used. Many machine learning models, particularly deep learning, function as "black boxes," offering predictions without explaining the decision-making

process. Doshi-Velez and Kim [15] provide a taxonomy of interpretability, distinguishing between application-grounded, human-grounded, and functionally-grounded approaches, clarifying when interpretability is necessary. Without transparency, candidates may distrust AI systems, perceiving them as unfair. Zhang et al. [16] found that public attitudes toward algorithmic hiring were generally negative, with demographic differences in perceptions.

Post-hoc explainers (LIME, SHAP, Integrated Gradients, and others) attribute predictions to input features; Section 2.5 states which method we use and why.

XAI also helps identify and mitigate biases in recruitment models. If a model is trained on biased data, it may prioritize certain features, such as gendered language or regional vocabulary. XAI enables auditors to see which tokens drive predictions before choosing a debiasing intervention. However, Novielli et al. [17] notes that fairness constraints may require revealing sensitive data, affecting privacy.

Despite these benefits, achieving full transparency in complex models is challenging. Deep networks in particular are hard to interpret end to end. For fairness auditing, XAI is most useful for showing which tokens and features drive group-level disparities before any debiasing step is chosen. Post-hoc explanations can support review. Group fairness metrics still decide whether disparity is acceptable.

2.2 Fairness in AI Recruitment Systems

Defining fairness in AI recruitment is complex and multifaceted. Rigotti [18] states fairness is "subjective," varying between applicants and HR officers. As AI

becomes prevalent in recruitment, ensuring these systems are fair and transparent is crucial. AI's potential to perpetuate biases inherent in historical data makes fairness a critical concern. Köchling & Wehner [19] reviewed discrimination in HR analytics, including screening tools, highlighting the need for fair AI systems.

Fairness lacks a universal definition applicable to all AI systems. Mehrabi et al. [10] note that while numerous fairness metrics exist, no single one is universally accepted. Kusner et al. [20] define fairness using counterfactuals: a decision is fair if it remains the same in an alternate world where the individual belongs to a different demographic group.

In recruitment, fairness ensures candidates are evaluated equitably, free from discriminatory factors like gender, race, or age. Fairness in AI systems can be categorized into group fairness, individual fairness, and procedural fairness, each addressing different aspects of fairness.

Fairness research can be divided into four main eras:

- **2018–2020:** classical TF–IDF/CV + early dual-encoders; first fairness metrics; post-processing (thresholding) dominates.
- **2021–2022:** BERT/DeBERTa/Roberta dual-encoders; cross-encoder reranking; better calibration; rising concern about proxy features.
- **2023–2024:** DRO/adversarial debiasing enter practice; counterfactual evaluation (ITS/ITPE); interpretable-by-design advocacy.
- **2025:** segment-specialist ensembles (MSLEF), long-context encoders, LLM-assisted auditing; fairness integrated into training objectives.

2.2.1 Group Fairness vs. Individual Fairness

Group Fairness

Group fairness ensures equitable treatment of demographic groups. As explained by González Sendino et al. [3], it ensures equal treatment for privileged and unprivileged groups in terms of outcomes. In recruitment, group fairness often involves equal hiring outcomes across demographic groups. Demographic parity, a widely used metric, ensures equal likelihood of being selected for interviews, regardless of qualifications. However, demographic parity can compromise model performance, as it may favor underrepresented but less qualified groups, leading to the fairness-accuracy trade-off.

Individual Fairness

Individual fairness ensures that similar candidates are treated similarly. Dwork et al. [21] introduce this concept, focusing on fairness at the individual level. Similar candidates should receive similar outcomes, regardless of race or gender. Counterfactual fairness [20] evaluates whether a candidate's outcome would differ if demographic characteristics changed, holding other factors constant. However, defining "similar" candidates consistently remains a challenge.

2.2.2 Other Fairness Definitions (Background)

Calibration asks whether predicted probabilities match outcomes within groups [4]. The four-fifths rule flags adverse impact when a group's selection rate falls below 80% of the highest group's rate. Fairness through unawareness drops protected attributes from inputs [22], yet proxy features in resume text can still

skew outcomes. We do not optimise these criteria in the experiments; the audit centres on robust TPR gaps (Section 3.5.3), GroupDRO, and city-swap wording robustness.

2.2.3 Fairness-Accuracy Trade-Offs

The fairness-accuracy trade-off is a challenge in AI recruitment. Fairness metrics, like demographic parity, can reduce model accuracy, especially with imbalanced datasets. Achieving fairness may require selecting less qualified candidates, impacting the model's performance. Researchers have developed techniques, such as fairness constraints and post-processing adjustments, to balance fairness and accuracy. Li, Raymond & Bergman [23] additionally note that ranking systems must balance exploiting known good candidates with exploring underrepresented talent, which interacts with fairness goals when historical data are skewed.

2.2.4 Fair Determination Trade-off

Kleinberg, Mullainathan & Raghavan [4] prove that calibration within groups, balance for the positive and negative classes cannot be simultaneously satisfied unless in specific cases. This formalizes the trade-offs in fairness debates, where developers must choose which errors to tolerate.

2.2.5 Demographic Parity and Equalized Odds

Demographic parity ensures hiring outcomes are equally distributed across demographic groups but often neglects qualification differences. Equalized odds,

conversely, ensures fairness by accounting for candidate qualifications while maintaining equal chances across groups. Both metrics are used in recruitment AI, with varying results.

Link to our group-fairness metric. Hardt et al. [24] define **equal opportunity** as equal true positive rates (TPR) across groups when the true label is positive. Our primary fairness statistic is a robust worst-case TPR gap across city (and secondarily gender and age) groups within each profession class, with a minimum support threshold (Chapter 3.5). This is closer to equal opportunity than to demographic parity on acceptance rates. We compare in-training mitigations under a fixed argmax decision rule rather than applying Hardt et al.’s group-specific score thresholds on every checkpoint.

2.2.6 Disparate Impact

Disparate impact occurs when AI systems unintentionally disadvantage certain demographic groups, even if sensitive features are not considered. For example, resume ranking systems may favor skills more common in one demographic, causing disparate impact. Machine learning algorithms can perpetuate such biases if not carefully monitored.

2.2.7 Ethical and Legal Considerations of Fairness

AI recruitment systems raise ethical and legal concerns, as biased systems can perpetuate discrimination. Mehrabi et al. [5] distinguish between “formal fairness” and “perceived fairness,” addressing legal and ethical issues. AI systems must comply with anti-discrimination laws and ensure transparency to main-

tain fairness. Feldman et al. [25] suggest further research on legal frameworks. Human-in-the-Loop (HITL) systems, where human recruiters validate AI recommendations, can mitigate bias but may slow the hiring process and introduce new biases.

2.3 Bias Mitigation in Recruitment AI

Bias in AI recruitment systems is well-documented, with many studies showing that biased training data can lead to unfair outcomes. These biases, if unchecked, can perpetuate discrimination in hiring, reflecting societal biases such as gender, racial, or age biases. As AI plays a central role in recruitment, addressing bias is crucial to ensure fairness and equality in hiring practices.

AI recruitment systems, such as resume screening and candidate ranking, are susceptible to biases if trained on biased datasets. Bias mitigation must occur at every stage of development, including training data, model training, and post-processing.

2.3.1 Bias in Training Data

A major source of bias in recruitment AI comes from the training data, which often reflects historical hiring biases. If training data shows a preference for certain groups, the AI system will likely replicate these biases in predictions. For example, gendered language in resumes can lead to gender bias, and disparities in technical roles can result in models favoring male candidates. Real resume corpora used in teaching and research settings, such as anonymized HeadHunter exports [11], inherit geographic and occupational imbalance that can surface as proxy bias in text classifiers. Mitigating bias requires diverse, inclusive training data, but achieving

a perfectly unbiased dataset is challenging.

2.3.2 Bias Mitigation Techniques

Once biases are identified in the training data, various techniques can mitigate their impact, categorized into preprocessing, in-processing, and post-processing methods [9]. Caton and Haas [10] note that selecting the appropriate approach depends on the scenario. Our experiments follow this taxonomy: inverse square-root reweighting and weighted city sampling (preprocessing); label smoothing, R-Drop, focal and class-balanced losses, GroupDRO, and adversarial debiasing (in-processing); training-time and post-hoc logit adjustment (post-processing); plus IG-guided token scrubbing as an input-level intervention related to proxy removal [26], [27].

Preprocessing Methods

Preprocessing adjusts the training data before model input. De-biasing input features, such as removing or neutralizing sensitive demographic data, prevents the AI from learning biases. Reweighting training data to ensure underrepresented groups are fairly represented also helps. However, overcompensating for certain groups can lead to overfitting, which reduces model performance.

Adversarial Debiasing

Adversarial debiasing uses a secondary model, the "adversary," to counteract discriminatory patterns during training. The primary model is trained to make accurate predictions, while the adversarial model attempts to predict sensitive attributes, like gender. The goal is for the primary model to make predictions without

relying on demographic features. However, this method can reduce model accuracy, presenting a trade-off between fairness and predictive power.

2.3.3 Post-Processing Methods

Post-processing methods adjust model outputs to ensure fairness. Re-ranking results ensures equal representation across demographic groups in recruitment. Thresholding adjusts the decision threshold for different groups, preventing unfair disadvantages. While post-processing can reduce bias, it does not address underlying causes, and is most effective when combined with preprocessing or in-processing techniques. Seppälä & Małecka [28] claim that preprocessing and in-processing outperform post-processing. Rudin et. al [29], [30] argue against relying on post-hoc explanations for black box models, emphasizing that such explanations can be misleading and reinforce false trust. Instead, they advocate designing models that are interpretable by construction.

2.3.4 Challenges and Trade-Offs in Bias Mitigation

Bias mitigation techniques come with challenges. The fairness-accuracy trade-off is significant, as improving fairness often reduces model accuracy. This is particularly problematic in high-stakes recruitment scenarios. Techniques like adversarial debiasing also increase model complexity, requiring additional resources and time. Legal and ethical concerns arise, especially when removing demographic data, as it may compromise transparency and accountability in decision-making. Balancing fairness with practical feasibility remains a key challenge in AI recruitment systems.

2.4 Proposed Solutions and Architectures

2.4.1 Distributionally Robust Optimization (DRO) for Fairness

Hashimoto et al. [31] train models to minimise worst-case loss over uncertain group distributions. GroupDRO [31] is the variant we implement: it reweights per-city training losses using explicit `city_group` labels (Chapter 3.7) to target geographic disparity from the baseline audit.

2.4.2 Adversarial Debiasing (Literature)

Zhang et al. [32] train a main predictor while an adversary tries to recover a protected attribute from its representations; the main model is updated to keep task accuracy and hide that signal. We run one exploratory gradient-reversal trial on 41 city groups (Chapter 4.2); Section 5.4 reports the failure mode.

2.4.3 Other Architectures and Tasks (Background)

FAIM [33] couples rule-based interpretability with fairness losses; we cite it as evidence that transparency can be built into the model class, while our audit uses IG on a BERT classifier before debiasing.

Du et al. [26] recommend diagnosing proxy bias with attributions, then applying pre-, in-, or post-processing; our five-phase protocol follows that pattern with IG, GroupDRO, scrubbing, and post-hoc logit adjustment.

MSLEF [34] fine-tunes segment-specific encoders with fairness terms in the objective; we use one full-text BERT encoder but the same audit-then-mitigate

logic at token level.

Resume2Vec [35] scores resume–job similarity in embedding space; our task is nine-way profession classification on Russian resumes [11].

Fairness in summarization [36] (FairExtract, FairGPT) balances summary quality with group coverage; the setting differs, but the multi-objective idea parallels our three-axis reporting.

2.4.4 Equality of Opportunity in Supervised Learning

Hardt et al. [24] define equalized odds (equal TPR and FPR across groups given the true label) and equal opportunity (equal TPR only). Their post-processing adjusts group-specific thresholds without retraining. We test a lightweight variant, post-hoc logit adjustment on the best label-smoothing checkpoint, without full per-group thresholds (Chapter 3.7).

2.4.5 Mitigating Demographic Bias in AI-based Resume Filtering

Deshpande et al. [27] match 135 Singapore finance resumes to job postings with tf–idf cosine similarity; baseline retrieval favours Malaysian and Indian candidates over Chinese (Table 2.1), consistent with nationality clusters in embedding space.

Fair-tf–idf. Terms receive an extra demographic *p-ratio* weight on top of tf–idf; a sigmoid variant trades fairness against accuracy [27]. Results on their matching task are summarised in Table 2.1.

Our IG-guided **data scrubbing** is conceptually related: demographically salient *tokens* are masked instead of tf–idf reweighted, with the same goal of re-

TABLE 2.1
Fair-tf-idf results on the resume matching task

Method	Fairness $\uparrow$	Accuracy (%) $\uparrow$
tf-idf	0.0723	90.00
fair-tf-idf	0.3673	73.33
fair-tf-idf + Sigmoid	0.9230	86.66

ducing reliance on proxy vocabulary [26], [27].

2.4.6 Identity Term Sampling and Geographic Counterfactuals

Sobhani and Delany [37] label weak gender from identity-term counts (ITS) or syntactic patterns (ITPE), then run group metrics and counterfactual edits on gendered surface forms. Our **city-swap** test uses the same idea for geographic tokens in Russian resumes: rewrite city and region names and measure prediction flips (Chapter 3.5.4) [20], [37].

2.5 Explainable AI Methods Used in This Study

Several post-hoc explainers are used in fairness work: LIME [38] fits a local linear surrogate around each input; SHAP [39] allocates Shapley-style feature contributions but scales poorly with long vocabularies; Grad-CAM [40] and DeepLIFT [41] target vision-centric or layerwise setups and are a weak fit for plain resume text. None of these are run in our experiments.

2.5.1 Integrated Gradients

Integrated Gradients [26], [42] integrate gradients from a baseline input to the actual resume along the path to the predicted class logit, yielding one attribution score per token (full definition in Chapter 3.6). IG applies directly to BERT subword inputs, is cheaper than SHAP at our sequence length, and is more stable than LIME’s local surrogate on nearby resume paraphrases.

We use IG in Phase 1 to flag city and age tokens, build the scrubbing and city-swap lexicon, and check alignment with group TPR gaps (P2). Explanations support the audit; group metrics remain decisive.

2.6 Research Orientation and Evaluation Framework

The literature above points to three takeaways for our study.

First, fairness in recruitment AI is multi-dimensional. Group fairness, individual fairness, calibration, and robustness to wording edits often pull in different directions [4], [5]. Second, explainability tools such as Integrated Gradients work best as diagnostics: they help locate suspicious features before mitigation [26], [29], [30]. Third, debiasing ideas that work for simple protected attributes (e.g., binary gender) often struggle on high-cardinality geography and on resume text with strong proxy features [27], [31], [32].

We evaluate each trained configuration on three separate axes (Chapter 3.4): macro-averaged F1 and accuracy (utility); robust worst-case and macro TPR gaps, mainly by `city_group`, with gender and age reported in Phase 1 for comparison (group fairness); and city-swap flip rate on a named subset of checkpoints

(wording robustness). One number cannot stand for overall fairness [4], [10]. The experiments map trade-offs across methods rather than crown one winner.

We adopt the following working propositions for the empirical chapters:

1. **P1 (Bias inheritance).** A BERT-based resume classifier fine-tuned on real Russian IT hiring data will exhibit measurable group disparities, with geographic attributes producing larger robust TPR gaps than gender or age.
2. **P2 (Diagnostic linkage).** Integrated Gradients attributions for city and age tokens will align directionally with observed group-level disparities, supporting attribution-guided input interventions (data scrubbing, city-swap evaluation).
3. **P3 (Conditional mitigation).** Debiasing methods will move some evaluation axes and leave others flat or worse: group-robust training may cut worst-case TPR gap at an F1 cost; regularizers may keep utility with modest gap shifts; scrubbing may cut flip rate without closing the aggregate TPR gap.
4. **P4 (No universal optimum).** No single configuration among the tested loss modifications, group-robust methods, input interventions, and combinations will dominate simultaneously on macro F1, worst-case TPR gap, and city-swap flip rate.

Together, P1–P4 define an audit-first agenda: measure disparities, explain salient tokens, try targeted mitigations, and record where each method lands on utility, gap, and flip rate. The chapters that follow test these expectations on our dataset and turn the results into deployment guidance.

Chapter 3

Methodology

This chapter lays out the *five-phase audit protocol* from Chapter 1. We define the nine-class task and HeadHunter dataset, the BERT baseline with city reweighting, and the three-axis evaluation setup (utility, group fairness, wording robustness). We then specify fairness metrics with geography as the main disparity target, Integrated Gradients for Phase 1, the debiasing families in Table 3.1, and the phase plan used in Chapter 4.

3.1 Task Definition

We formulate the resume screening problem as a multi-class text classification task. Given a resume represented as a sequence of tokens $x = (x_1, x_2, \dots, x_n)$, the objective is to predict the professional category $\hat{y}$ from a predefined set of K classes:

$$y \in \mathcal{Y} = \{c_1, c_2, \dots, c_K\} \quad (3.1)$$

where $K = 9$ categories cover the main segments of the IT labor market. The predicted label $\hat{y}$ is obtained by applying a learned function f_θ to the input sequence,

parameterized by a deep neural network with weights θ .

Each resume in our dataset is additionally annotated with categorical sensitive attributes: geographic location $g_{\text{city}} \in \mathcal{G}_{\text{city}}$, gender $g_{\text{gender}} \in \{\text{Male}, \text{Female}\}$, and age group $g_{\text{age}} \in \mathcal{G}_{\text{age}}$. These attributes are not used as model inputs during training. They are used only after training to group test predictions for fairness evaluation. The distinction between task-relevant features (skills, experience, education) and potentially sensitive attributes (location, gender, age) is central to the audit design.

The nine professional categories used as target classes are:

1. `backend_general_dev`: backend and general software development
2. `web_frontend`: web frontend development
3. `sysadmin_devops_network`: system administration, DevOps, and networking
4. `project_product`: project and product management
5. `sales_account`: sales and account management
6. `tech_support_helpdesk`: technical support and helpdesk
7. `it_governance_leadership`: IT governance and leadership
8. `technical_specialized`: specialized technical roles
9. `generic_it_ops`: general IT operations

This categorization was derived from occupation fields in anonymized resumes exported from HeadHunter (hh.ru) [11]. While the categories are not exhaustive, they represent the major employment clusters in the Russian IT sector and provide a sufficiently granular target space for fairness analysis.

3.2 Dataset Description

3.2.1 Data Provenance and Preprocessing

Resumes originate from the public HeadHunter (hh.ru) coursework dataset and analysis project *Project 1: Resume Analysis on HeadHunter* [11] (44,744 anonymized rows in the raw release; 27,550 after our IT-focused filtering). The source is curated teaching material, not a peer-reviewed dataset publication.

External validity. The results describe one Russian IT hiring corpus. Label rules, city grouping, class balance, and labour-market structure may differ elsewhere. We publish splits and metrics so others can replicate or compare on other HR datasets. Without that, claims beyond hh.ru-style IT resumes stay tentative.

After removing incomplete records, restricting to IT-relevant occupations, and mapping fine-grained labels to the nine supercategories in Section 3.1, sensitive attributes (city, gender, age group) are taken from resume metadata and used only for post-hoc fairness grouping, not as model inputs.

Each sample contains the resume text plus metadata fields. The dataset was partitioned into training (16,530 samples, 60%), validation (5,510 samples, 20%), and test (5,510 samples, 20%) subsets using stratified random sampling to preserve the class distribution across splits.

The text of each resume is written in Russian and varies in length, formatting, and level of detail. Resume texts were truncated or padded to a maximum sequence length of 128 tokens after tokenization with the BERT WordPiece tokenizer. This length balances context and computational cost for BERT fine-tuning.

3.2.2 Attribute Distribution

Before choosing the main sensitive attribute for debiasing experiments, we analyzed how applicants are distributed across groups.

Geographic distribution. The dataset spans 41 city groups. Moscow and Saint Petersburg together account for 44.6% of samples (12,284 resumes); the remaining 39 cities share 55.4% (Figure 3.1). Many regional cities contain fewer than 50 resumes each. Small groups make per-group metrics noisy; we address this with a support filter in Section 3.5.2. Geographic imbalance increases the risk that the model learns spurious links between regional vocabulary and job category.

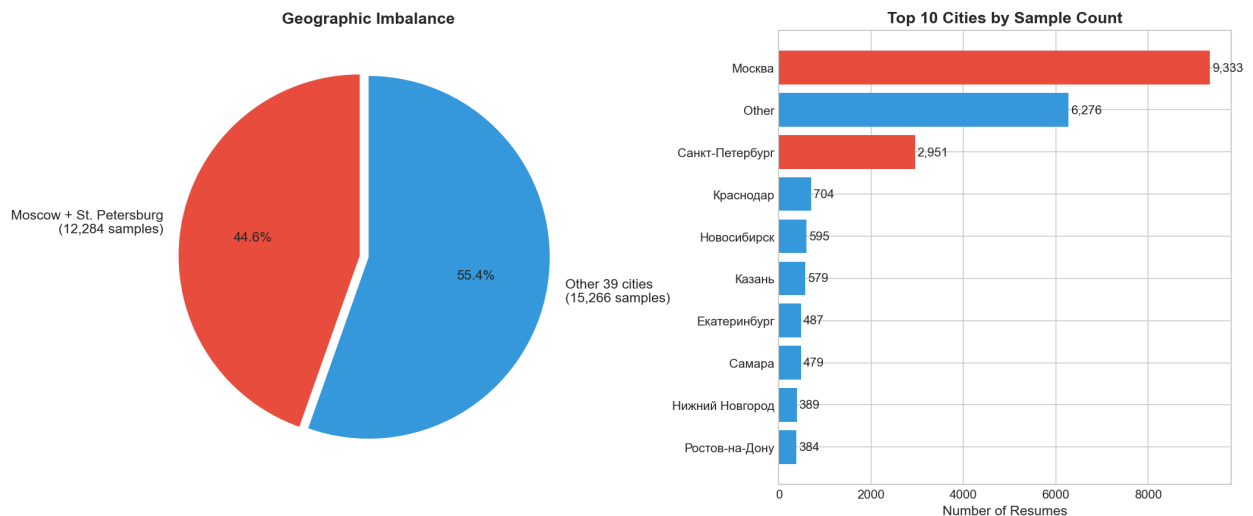


Figure 3.1. Geographic imbalance in the filtered dataset: share of resumes from Moscow and Saint Petersburg versus other cities (left), and the ten largest city groups (right).

Labels and age. The nine IT supercategories are imbalanced (e.g., SysAdmin/DevOps and Generic IT dominate). Age is grouped into seven bands; the known-age mass concentrates in 26–35, while a large *Unknown* band reflects missing or coarse metadata (Figure 3.2). Gender is recorded as Male/Female with an

approximately 70%/30% split.

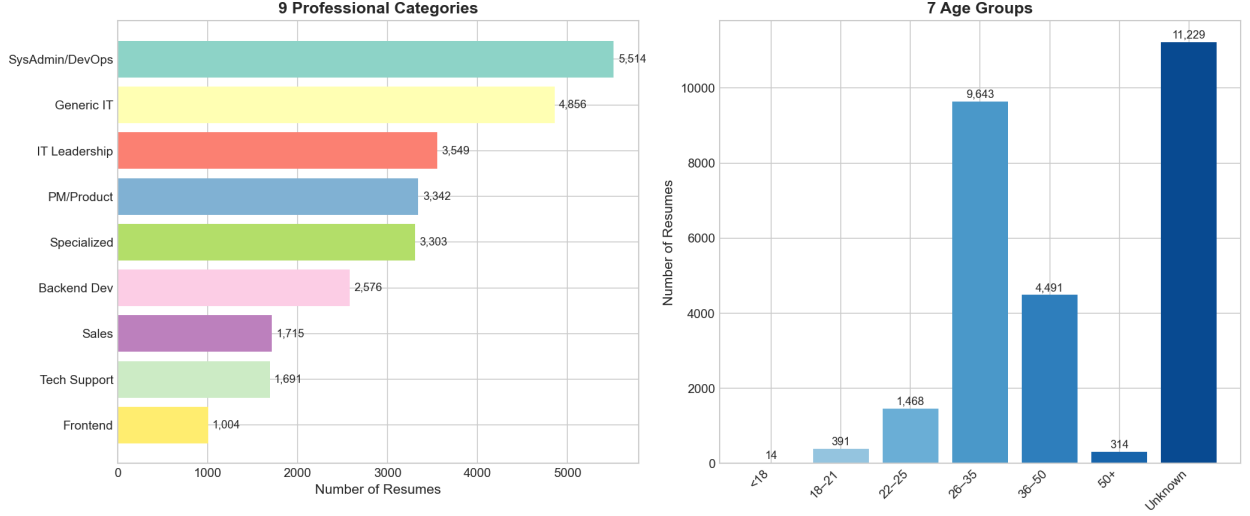


Figure 3.2. Class distribution over nine professional categories (left) and resume counts by age group (right).

Focus on geography. Chapter 4 compares worst-case TPR gaps across city, gender, and age on the baseline model. Geographic groups show the largest disparity; primary debiasing experiments therefore target city-related bias, while gender and age gaps are reported for context.

3.3 Baseline Model and Training Setup

Our classification model is `bert-base-uncased` (12 layers, 12 heads, hidden size 768; about 110M parameters). All debiasing configurations share this backbone unless noted otherwise. The classifier uses the `[CLS]` token representation from the final layer:

$$\hat{y} = \text{softmax}(W \cdot h_{[\text{CLS}]} + b) \quad (3.2)$$

where $W \in \mathbb{R}^{K \times 768}$ and $b \in \mathbb{R}^K$ are learnable parameters. All configurations start from this architecture unless noted otherwise.

3.3.1 Inverse Square-Root Sample Reweighting

To reduce the impact of severe group imbalance (Section 3.2.2), each training sample receives a weight inversely proportional to the square root of its city-group count:

$$w_i = \frac{1}{\sqrt{n_{g_i}}} \cdot \frac{1}{Z} \quad (3.3)$$

where n_{g_i} is the number of training samples in the same city group as sample i , and Z normalizes weights so their sum equals the number of training samples. This scheme upweights small cities without the extreme weights of inverse-frequency reweighting.

3.3.2 Training Hyperparameters

Unless stated otherwise, all models use:

- Pre-trained model: BERT-base-uncased
- Learning rate: 2×10^{-5} with linear warmup over the first 10% of steps
- Batch size: 8
- Maximum sequence length: 128 tokens
- Training epochs: 2
- Primary loss: cross-entropy (replaced or extended for specific debiasing methods)

Training runs use random seed 42 for the Phase 2–3 hyperparameter search. Phase 4 retrains three anchor configurations under seeds 7, 13, and 21 (Section 4.4.1). Checkpoints are chosen by lowest validation loss; all reported metrics are computed on the held-out test set.

3.3.3 Search Depth and Random Seeds

The Phase 2–3 grid is wide rather than exhaustive: one tuned checkpoint per family on seed 42, chosen by validation loss. That lets us map a trade-off surface, but fairness rankings (e.g., label smoothing vs. R-Drop on worst-case gap) can shift with the seed. Phase 4 retrains R-Drop, label smoothing, and the weighted city sampler on seeds $\{7, 13, 21\}$ (Table 4.3). Other configurations should be read as single-seed illustrations unless re-checked. If fairness gates release, report mean $\pm$ std over several seeds for the chosen model, not only seed 42.

3.4 Evaluation Framework

No single number tells us whether a resume classifier is acceptable for deployment. We evaluate each configuration on three axes and report them separately.

- **Utility:** accuracy and macro-averaged F1 on the test set.
- **Group fairness:** robust worst-case and macro-average TPR gaps (Section 3.5).
- **Wording robustness:** city-swap flip rate (Section 3.5.4) on a selected subset of checkpoints (Section 3.5.5). This is a wording stress test, not a full causal fairness proof.

A method may gain on one axis and lose on another. We map those trade-offs instead of hunting for one model that tops every metric. In Chapter 4, a missing flip value means city-swap was skipped for that run, not that the model never flips.

3.5 Fairness Metrics

3.5.1 True Positive Rate and TPR Gap

For each professional category c and demographic group g , we measure the true positive rate (TPR):

$$\text{TPR}_{c,g} = P(\hat{y} = c \mid y = c, G = g) \quad (3.4)$$

The TPR gap for category c is the difference between the highest and lowest group TPR:

$$\Delta \text{TPR}_c = \max_{g \in \mathcal{G}} \text{TPR}_{c,g} - \min_{g \in \mathcal{G}} \text{TPR}_{c,g} \quad (3.5)$$

We report the *worst-case gap* $\max_c \Delta \text{TPR}_c$ across categories and the *macro-average gap* (mean of ΔTPR_c). Lower values mean smaller spread between groups. The metric follows **equal opportunity**: similar true positive rates across groups when the true label is positive [24]. We do not use demographic parity on raw acceptance rates as the main score. Figure 3.3 shows one category across three city groups.

3.5.2 Robust Fairness Evaluation

TPR estimates for rare group–class pairs are unstable: a few misclassified resumes can move the rate from 0% to 100%. To avoid inflating gaps with noise,

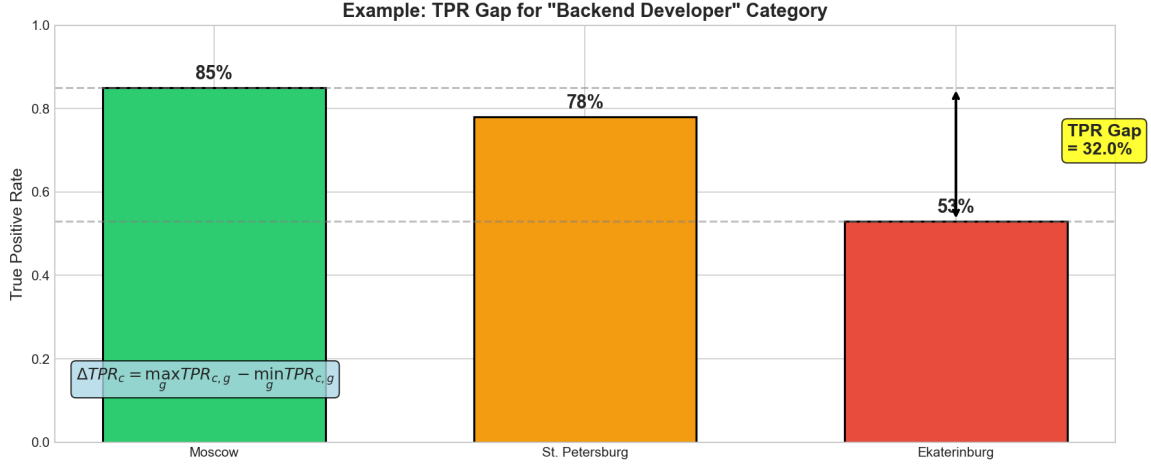


Figure 3.3. Example TPR gap for the `backend_general_dev` category across three cities. The gap is the difference between the highest and lowest group TPR (Equation 3.5).

we only include pairs with at least $\tau = 30$ test samples:

$$\Delta TPR_c^{\text{robust}} = \max_{\substack{g \in \mathcal{G} \\ n_{c,g} \geq \tau}} TPR_{c,g} - \min_{\substack{g \in \mathcal{G} \\ n_{c,g} \geq \tau}} TPR_{c,g} \quad (3.6)$$

where $n_{c,g}$ is the test count for class c and group g . Chapter 4 shows how much the robust filter changes the baseline gap compared with unfiltered metrics.

3.5.3 Why We Use TPR Gap

Fairness in hiring has many definitions [4], [5], [10]. We use **robust worst-case and macro-average TPR gaps** as the main group-fairness scores because:

1. The task asks whether people who truly belong to class c are recognised equally across city (and, in Phase 1, gender and age) groups. That matches **equal opportunity** [24].
2. Gaps split by class and group, so we can see which profession and region cells hurt most (Figure 3.3).

3. Many training fixes we test (GroupDRO, reweighting) already target group-conditional loss; TPR is the natural check after training.

We also report other ideas from the literature but do not treat them as primary targets: demographic parity of predicted labels [4]; full equalized odds with per-group thresholds [24] (we only try post-hoc logit adjustment on one model, Section 4.4.2); calibration by group; gender and age gaps in Phase 1 for context while debiasing focuses on `city_group` (Section 3.2.2, Chapter 4).

The thesis documents disparity in who gets the correct positive label under a fixed argmax rule. It does not claim legal fairness or prove causal non-discrimination.

3.5.4 City-Swap Counterfactual Test

To measure robustness to geographic wording, we apply a city-swap perturbation to test resumes. City-related tokens in the text are replaced with a target city name from a fixed list (Moscow, Ekaterinburg, Novosibirsk, Krasnodar, Voronezh). For models trained with data scrubbing, sensitive tokens are masked with `[MASK]` *after* the swap at inference time, matching training.

Given original input x and perturbed input $\tilde{x}$:

$$\text{flip}(x) = \mathbb{1}[f_{\theta}(x) \neq f_{\theta}(\tilde{x})] \quad (3.7)$$

The *overall flip rate* is the fraction of test resumes for which the predicted class changes under at least one target-city swap. We also report per-target-city flip rates.

The swap lexicon (about 70 tokens) is built from city names and age-related terms highlighted in the IG audit (Section 3.6).

How to read city-swap. City-swap replaces explicit city and region tokens (and related lexicon entries) while keeping skills and employment history as written. It checks whether predictions change when geographic wording changes. That is useful for spotting spurious location cues, especially with scrubbing (Table 4.2).

Rewriting a city can also change how plausible a resume looks (employer and location fit). We do not model full labour-market counterfactuals [20]. A low flip rate means low sensitivity to *this* edit protocol. Flip rate and TPR gap can move in opposite directions (Chapter 5, Section 5.2). We report both where city-swap was run.

3.5.5 Scope of City-Swap Evaluation

All trained configurations (>25 hyperparameter runs) receive utility and robust TPR-gap metrics on the full test set. **City-swap flip rates are not computed for every run:** the test is reserved for models that are either the best tuned variant within a debiasing method family or an explicitly chosen strong combined configuration from Phase 2–3.

We apply city-swap to:

- **Five family-representative single-method models:** Label Smoothing ($\epsilon = 0.1$), R-Drop ($\alpha = 1.0$), Class-Balanced CE ($\beta = 0.99$), GroupDRO ($\eta = 0.1$), and Focal Loss ($\gamma = 2$), each the best validation-loss checkpoint in its family;
- **Four primary combined models:** Data Scrubbing + Label Smoothing, Label Smoothing + R-Drop, Data Scrubbing + R-Drop, and Class-Balanced CE + Label Smoothing, chosen after strong accuracy and gap on validation;

- **Extended combined models** when needed for a stated hypothesis (Data Scrubbing + GroupDRO at $\eta \in \{0.05, 0.1\}$);
- **Post-hoc logit adjustment** applied to the best Label Smoothing model without retraining;
- the **reference baseline** (BERT with inverse square-root reweighting; flip rate 7.7%, Chapter 4.1.1).

Intermediate hyperparameter trials, dominated variants, and negative results (e.g., Adversarial Debiasing) are reported on accuracy and gap only. Cross-model flip comparisons therefore use this *common subset* of named configurations, not the full experimental inventory.

3.6 Integrated Gradients for Bias Audit

Before applying debiasing, we check whether the baseline model uses sensitive tokens and which tokens matter most. We use Integrated Gradients (IG) [42], a gradient-based attribution method.

IG assigns an attribution score to each token by integrating gradients along a path from a baseline input x' (zero embeddings) to the actual input x :

$$\text{IG}_i(x) = (x_i - x'_i) \times \int_{\alpha=0}^1 \frac{\partial F(x' + \alpha(x - x'))}{\partial x_i} d\alpha \quad (3.8)$$

where F is the logit for the predicted class. In practice we approximate the integral with $M = 50$ Riemann steps (Equation 3.9).

We compute mean attributions for city-name and age-related tokens across subgroups and categories (Figures 4.2–4.3). Tokens with large consistent attribu-

tions are added to the scrubbing and city-swap lexicon. Token-level patterns are compared with group-level TPR gaps in Chapter 4.1.3.

How to read IG. Integrated Gradients [26], [42] assign token-level scores from the trained classifier. We use them to build the scrubbing and city-swap lexicon and to see whether token patterns line up with group gaps (P2). Attributions can shift with paraphrases, baselines, and aggregation choices. They may miss document-level cues (employer names, layout) and subtle gender proxies in subwords. IG supports the audit; group TPR gaps and legal review still matter. Chapter 5 uses IG as diagnostic evidence alongside group metrics.

$$\text{IG}_i(x) \approx (x_i - x'_i) \times \frac{1}{M} \sum_{m=1}^M \frac{\partial F(x' + \frac{m}{M}(x - x'))}{\partial x_i} \quad (3.9)$$

3.7 Debiasing Methods Under Comparison

We test more than 25 training configurations built from the methods below. All use sqrt reweighting by city group unless noted. Hyperparameters are tuned on the validation set; Chapter 4 reports test-set results.

Coverage of methods. We compare many families with one tuned representative each on seed 42, plus selected stacks and one adversarial trial. This is a structured map of which ideas move utility, gap, or flip rate under a fixed training budget. It is not an exhaustive hyperparameter search for every method. Follow-up work can narrow the list and spend more seeds per family (Section 3.3.3).

GroupDRO. GroupDRO reweights per-city losses during training so that high-loss groups receive more weight [31]. Group weights are updated as $q_g^{(t+1)} \propto$

TABLE 3.1
Debiasing method families evaluated in this study

Family	Role in the study
Baseline + inverse sqrt reweighting	Reference point on the trade-off map
Label smoothing	Regularization; tune $\epsilon \in \{0.05, 0.1, 0.15\}$
R-Drop	Consistency regularization; tune $\alpha \in \{0.5, 1.0, 2.0\}$
Focal loss	Hard-example weighting; tune γ
Class-balanced CE / focal	Class rebalancing; tune β, γ
Logit adjustment	Prior correction in training and post-hoc
Weighted city sampler	Oversample small cities; $p \in \{0.5, 1.0, 1.5\}$
GroupDRO	Worst-group robust loss; tune $\eta \in \{0.05, 0.1, 0.2\}$
Data scrubbing	Mask sensitive tokens from IG lexicon
Adversarial debiasing	Gradient reversal on city group; $\lambda = 0.5$
Combined (primary)	Data Scrubbing + Label Smoothing; Label Smoothing + R-Drop; Data Scrubbing + R-Drop; Class-Balanced CE + Label Smoothing
Combined (extended)	Data Scrubbing + GroupDRO (Chapter 4)

$q_g^{(t)} \exp(\eta \mathcal{L}_g^{(t)})$ (Equation 3.10).

$$q_g^{(t+1)} \propto q_g^{(t)} \cdot \exp(\eta \cdot \mathcal{L}_g^{(t)}) \quad (3.10)$$

Data scrubbing. Sensitive tokens $t_i \in \mathcal{S}$ are replaced with [MASK] during training; scrubbing may also be applied at inference for combined models:

$$\tilde{x}_i = \begin{cases} [\text{MASK}] & \text{if } t_i \in \mathcal{S}, \\ x_i & \text{otherwise.} \end{cases} \quad (3.11)$$

Other loss modifications. Label smoothing, R-Drop, focal loss, class-balanced loss, and logit adjustment follow standard definitions (Chapter 2). They mainly change how the classifier is trained, not the input text.

Adversarial debiasing. An adversarial head predicts city group from [CLS] representations while the main model is trained to resist it (gradient reversal). This

tests whether representation-level hiding works when the protected attribute has many values (41 city groups).

3.8 Experimental Protocol

The **five-phase audit protocol** comprises the following stages. Phases 1–3 search for debiasing options; Phases 4–5 assess reproducibility and consolidate trade-offs for deployment.

Phase 1: Diagnostic audit. Baseline with inverse square-root reweighting; TPR gaps by city, gender, and age; Integrated Gradients and lexicon for scrubbing and city-swap (Chapter 4.1.3).

Phase 2: Debiasing benchmark. Train and tune all method families in Table 3.1 (>25 runs, seed 42); report accuracy, macro F1, and robust TPR gaps on the full test set.

Phase 3: Combinations and counterfactuals. Train primary and extended combined models; city-swap on the subset in Section 3.5.5.

Phase 4: Validation. *Reproducibility:* three anchor models (R-Drop, Label Smoothing, Weighted City Sampler) under seeds 7, 13, and 21 (Section 4.4.1). *Post-processing:* post-hoc logit adjustment on the best Label Smoothing model without retraining (Section 4.4.2).

Phase 5: Trade-off map. Consolidate Phases 1–4 into axis-specific optima, failure modes, and deployment recommendations (Sections 4.5 and Chapter 5).

Chapter 4

Experiments

This chapter reports results from the five-phase protocol in Section 3.8. Each configuration is scored on utility (accuracy, macro F1), group fairness (robust TPR gaps by city group, $\tau = 30$; gender and age only in Phase 1), and, for a selected subset, wording robustness (city-swap flip rate). We name the best option per axis and record failure modes; deployment guidance is in Chapter 5.

Metrics, data splits, training defaults, and city-swap scope are defined in Chapter 3. Unless noted, debiasing runs use seed 42, inverse square-root city reweighting, and checkpoints selected by validation loss; test-set metrics are reported for every trained configuration (>25 runs). Flip rates apply only to the subset in Section 3.5.5. A dash in Table 4.2 means city-swap was skipped for that row.

4.1 Phase 1: Diagnostic Audit

4.1.1 Baseline

The baseline (BERT + inverse square-root reweighting) reaches accuracy 0.609 and macro F1 0.621. Robust worst-case TPR gap is 0.329 (32.9%); macro-average gap is 0.116. The baseline city-swap flip rate is 7.7%: changing geographic wording often changes the predicted class, even when group-level gaps look moderate.

4.1.2 Fairness by Sensitive Attribute

Figure 4.1 compares robust worst-case TPR gaps on the baseline model. City groups show the largest disparity (32.9%), followed by age (18.2%) and gender (12.4%). Primary debiasing therefore targets geography. Without the $\tau = 30$ filter the baseline gap is 48.2%; roughly one third of that figure comes from rare group-class pairs rather than stable disparity.

4.1.3 Integrated Gradients

Integrated Gradients (Section 3.6) attribute baseline predictions to individual tokens. Figures 4.2 and 4.3 show mean attribution scores for lexicon entries matched in resume text, by profession (green positive, red negative).

Figure 4.4 shows local IG attributions on four misclassified test resumes: city names and domain wording among the strongest contributors to wrong profession labels.

Tokens with high $|IG|$ in Figures 4.2–4.3 were added to the scrubbing and

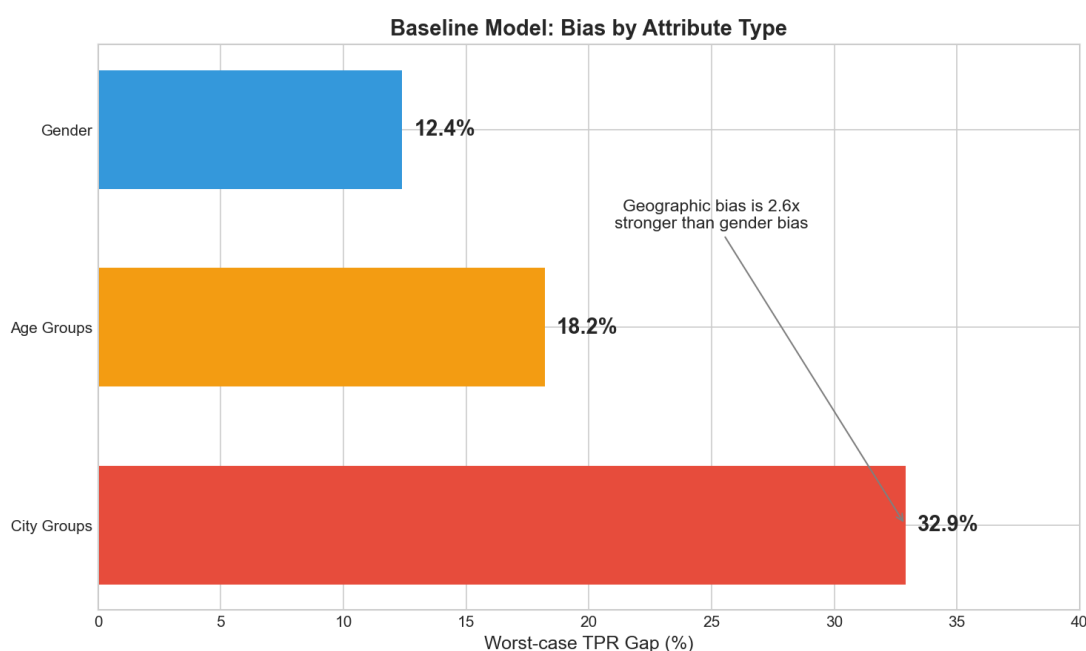


Figure 4.1. Baseline worst-case TPR gap by sensitive attribute (robust metric, $\tau = 30$). Geographic disparity exceeds age and gender.

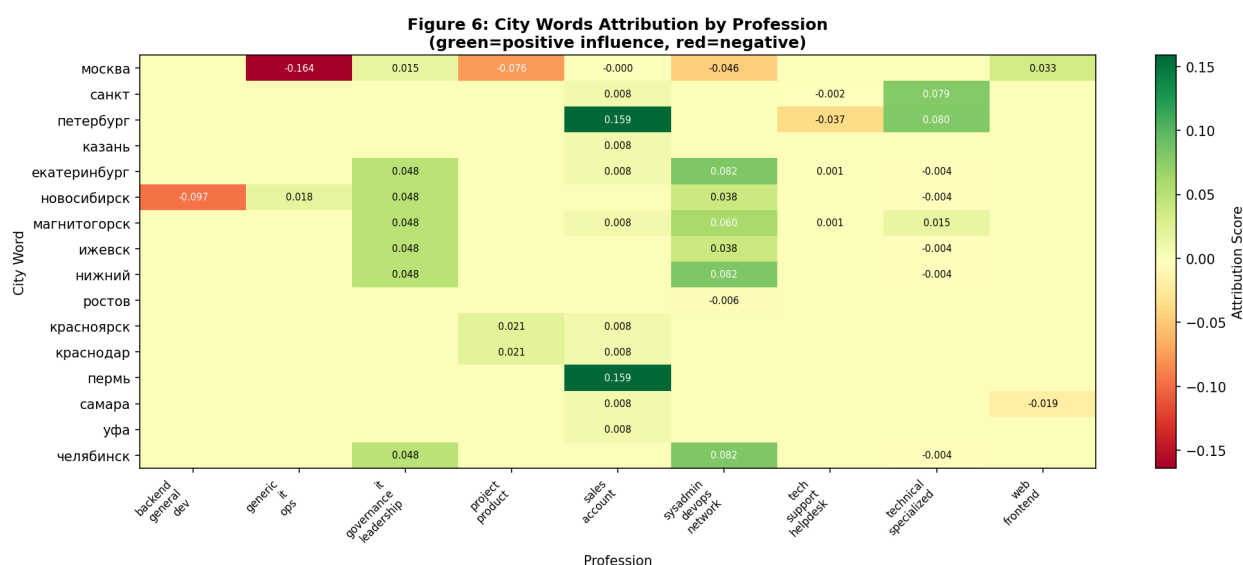


Figure 4.2. City-linked token attributions by profession on the baseline model. Strong positive scores for Saint Petersburg and Perm on sales_account; Moscow is negative for generic_it_ops (-0.164) and backend_general_dev.

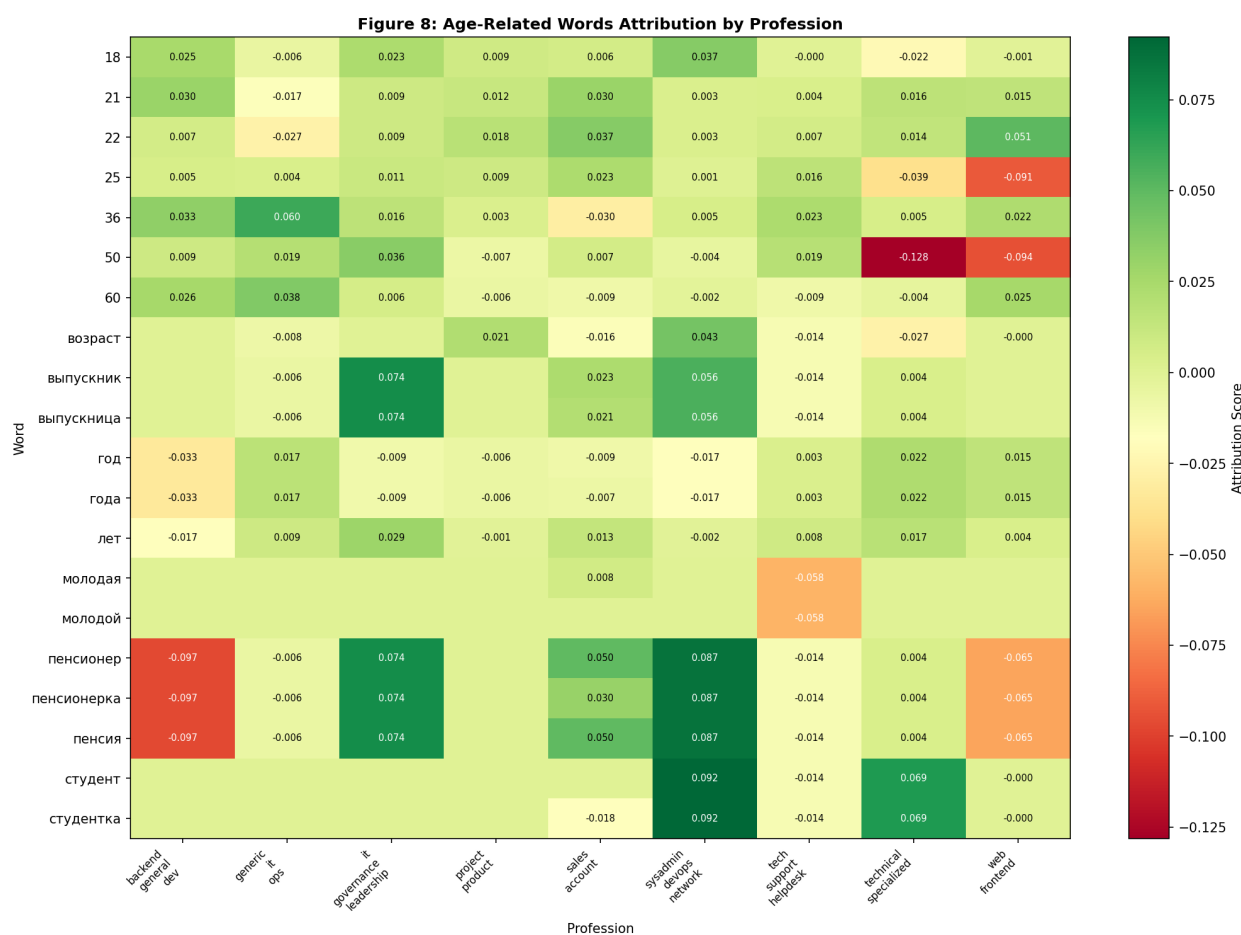


Figure 4.3. Age-related token attributions by profession. “пенсионер” (retired person) is negative for backend_general_dev (-0.097) but positive for it_governance_leadership ($+0.074$); “студент” (student) favors sysadmin_devops_network.

Figure 4: Profession Misclassification

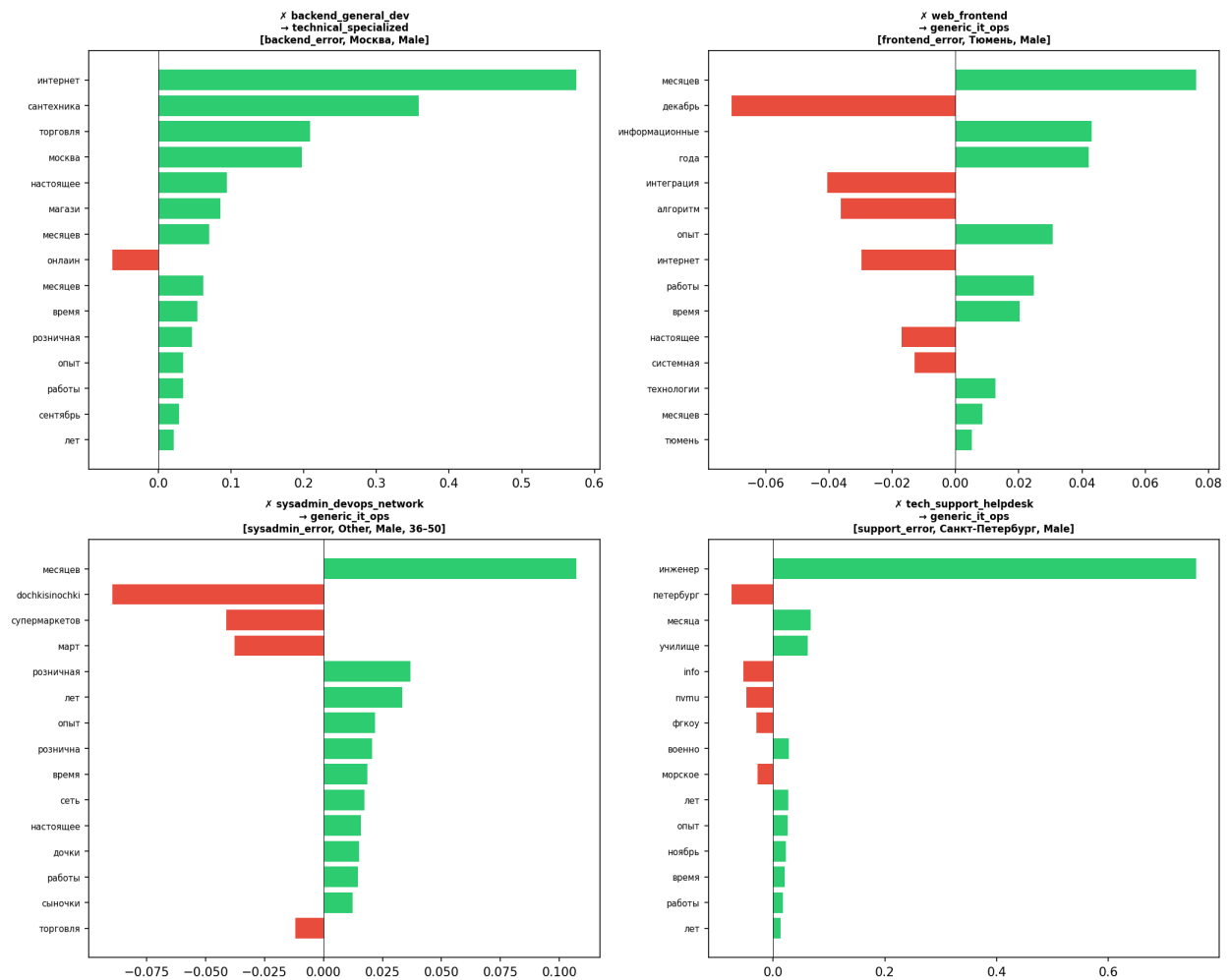


Figure 4.4. Integrated Gradients on selected baseline profession errors (green supports predicted class, red opposes). Examples include Moscow and Saint Petersburg tokens steering predictions away from the true category.

city-swap lexicon for Phases 2–3.

4.2 Phase 2: Debiasing Benchmark

Table 4.1 lists the best checkpoint per method family after hyperparameter tuning (seed 42). No configuration improves all metrics at once.

TABLE 4.1
Representative single-method models (seed 42, $\tau = 30$). Weighted City Sampler replaces sqrt reweighting

Model	Acc	Macro F1	Worst gap	Macro gap
Baseline	0.609	0.621	0.329	0.116
Label Smoothing ($\epsilon=0.1$)	0.608	0.623	0.318	0.136
R-Drop ($\alpha=1.0$)	0.607	0.620	0.312	0.127
Focal Loss ($\gamma=1$)	0.604	0.619	0.324	0.134
Class-Balanced CE ($\beta=0.99$)	0.607	0.621	0.388	0.137
Logit Adjustment ($\tau=0.5$)	0.599	0.617	0.371	0.126
Weighted City Sampler ($p=1.5$)	0.561	0.561	0.288	0.107
GroupDRO ($\eta=0.1$)	0.546	0.562	0.276	0.131
GroupDRO ($\eta=0.05$)	0.575	0.590	0.288	0.122

Utility-favorable methods. Label smoothing yields the highest macro F1 (0.623) with a small gap reduction (0.318). R-Drop matches baseline-level F1 (0.620) with the best gap among regularizers (0.312).

Class-imbalance and prior correction. Focal loss does not beat the baseline on gap; extended γ tuning confirms it targets hard *classes*, not city groups. Class-balanced CE worsens the gap to 0.388. Training-time logit adjustment ($\tau = 0.5$) reaches gap 0.371.

Group-robust methods. GroupDRO at $\eta = 0.1$ minimizes worst-case gap (0.276) at a large F1 cost (0.562). A smaller step ($\eta = 0.05$) trades some gap for F1 0.590. The weighted city sampler reaches gap 0.288 with F1 0.561, a similar utility cost.

Tuning. Within each family, hyperparameters shift positions on the frontier but do not remove the trade-off (e.g., label smoothing $\epsilon = 0.1$ beats $\epsilon \in \{0.05, 0.15\}$ on F1; R-Drop $\alpha = 2.0$ worsens gap to 0.335).

Exploratory: adversarial debiasing. One gradient-reversal run with city group as the adversarial target ($\lambda = 0.5$, 41 groups) was evaluated outside the main tuning grid: accuracy 0.574, macro F1 0.594, worst-case gap 0.429 (42.9%), below the baseline on both utility and fairness. City-swap was not run. Chapter 5.4 discusses implications.

4.3 Phase 3: Combinations and Counterfactuals

Four primary combined models and one extended pair were trained on seed 42. Table 4.2 reports classification metrics and city-swap flip rates for the evaluated subset (Section 3.5.5).

None of the combinations beat single-method best gaps (R-Drop 0.312; GroupDRO 0.276). **Data Scrubbing + R-Drop** gives the highest F1 (0.622) and lowest flip (0.09%), two orders of magnitude below the baseline. Scrubbing-based stacks keep flip near zero while loss-only stacks (Label Smoothing + R-Drop, Class-Balanced CE + Label Smoothing) stay near baseline flip (6–9%).

Data Scrubbing + GroupDRO. At $\eta = 0.1$, F1 is 0.588 and gap 0.324, below GroupDRO alone (F1 0.562, gap 0.276). At $\eta = 0.05$, F1 rises to 0.601 but gap to 0.359. City-swap flip stays negligible (0.00% and 0.15%): scrubbing calms token-level sensitivity even when group gaps stay high.

Stacks can improve flip rate without closing the TPR gap. Low flip and low group disparity are separate outcomes.

TABLE 4.2
City-swap subset: classification and overall flip rate (%). Scrubbing models mask sensitive tokens after swap at inference

Model	F1	Worst gap	Flip (%)
<i>Reference</i>			
Baseline (legacy eval)	0.621	0.329	7.7
<i>Family representatives (single-method)</i>			
Label Smoothing ($\epsilon=0.1$)	0.623	0.318	8.9
R-Drop ($\alpha=1.0$)	0.620	0.312	6.2
Class-Balanced CE ($\beta=0.99$)	0.621	0.388	5.8
GroupDRO ($\eta=0.05$)	0.590	0.288	8.7
Focal Loss ($\gamma=1$)	0.619	0.324	8.7
<i>Primary combined models</i>			
Data Scrubbing + Label Smoothing	0.620	0.324	0.16
Label Smoothing + R-Drop	0.620	0.335	6.0
Data Scrubbing + R-Drop	0.622	0.324	0.09
Class-Balanced CE + Label Smoothing	0.617	0.341	8.6
<i>Extended combined</i>			
Data Scrubbing + GroupDRO ($\eta=0.1$)	0.588	0.324	0.00
Data Scrubbing + GroupDRO ($\eta=0.05$)	0.601	0.359	0.15
<i>Post-hoc (no retrain)</i>			
Post-hoc logit adj. on Label Smoothing ($\tau=0.5$)	0.624	0.347	7.0

4.4 Phase 4: Validation

4.4.1 Training Reproducibility

Three anchor models (R-Drop, Label Smoothing, and Weighted City Sampler) were retrained with seeds 7, 13, and 21 (Table 4.3). Metrics are mean $\pm$ std over those seeds; seed 42 results in Table 4.1 are shown for comparison.

TABLE 4.3
Multi-seed stability (seeds 7, 13, 21)

Model	Macro F1	Worst gap	Seed-42 gap
R-Drop ($\alpha=1.0$)	0.617 ± 0.005	0.329 ± 0.010	0.312
Label Smoothing ($\epsilon=0.1$)	0.610 ± 0.006	0.312 ± 0.044	0.318
Weighted Sampler ($p=1.5$)	0.566 ± 0.005	0.322 ± 0.048	0.288

R-Drop has the most stable F1 (± 0.005). Label smoothing shows gap variance ± 0.044 (one seed reached 0.271, others near 0.329). Fairness gains from a single seed should not be over-interpreted; for stable deployment, R-Drop is preferable when gap and F1 must move together.

4.4.2 Post-Processing Limit

Post-hoc logit adjustment on the Label Smoothing checkpoint ($\tau \in \{0.3, 0.5, 0.7\}$, no retraining) raises macro F1 to 0.626 at $\tau = 0.3$ but worsens worst-case gap to 0.341–0.347. At $\tau = 0.5$, flip rate is 7.0% (Table 4.2), close to the baseline. Adjusting scores after training did not fix geographic gaps on this data.

4.5 Phase 5: Trade-Off Map

Table 4.4 summarizes axis-specific choices; there are no row wins on all metrics.

TABLE 4.4
Best configuration by deployment priority (seed 42 unless noted)

Priority	Model	Macro F1	Gap / flip
Highest F1	Label Smoothing ($\epsilon=0.1$)	0.623	gap 0.318
Balanced F1 + gap	R-Drop ($\alpha=1.0$)	0.620	gap 0.312
Lowest worst-case gap	GroupDRO ($\eta=0.1$)	0.562	gap 0.276
Lowest city-swap flip	Data Scrubbing + GroupDRO ($\eta=0.1$)	0.588	flip 0.00%
Best F1 with low flip	Data Scrubbing + R-Drop	0.622	flip 0.09%
Stable across seeds	R-Drop ($\alpha=1.0$, 3 seeds)	0.617	gap 0.329 $\pm$ 0.010

Improving worst-case gap from 0.318 to 0.276 costs about 6 F1 points. Word-ing robustness (flip rate) can be near-zero (scrubbing combinations) without closing the TPR gap.

Chapter 5

Evaluation and Discussion

Chapter 4 reports the test-set numbers: accuracy, macro F1, robust TPR gaps ($\tau = 30$), and city-swap flip rates on a named subset (Table A.1 lists every checkpoint). This chapter ties those results to the research questions (Section 1.6), propositions P1–P4 (Section 1.7), and deployment. Failed runs and seed-sensitive rankings belong in the main narrative.

5.1 Answers to Research Questions

RQ1: Measurable bias across demographic groups. Yes. On the baseline (Section 4.1.1), city groups show the largest robust worst-case TPR gap (32.9%), above age (18.2%) and gender (12.4%) in Figure 4.1. The baseline already uses inverse square-root reweighting by city; the 32.9% gap shows that reweighting alone leaves geographic disparity on this data. That supports P1 and the idea that bias comes from historical hiring patterns and proxy wording in resumes [3], including more than explicit city names.

RQ2: Integrated Gradients as audit evidence. IG attributions on baseline errors (Section 4.1.3) line up with group-level gaps: city and age tokens get large $|IG|$ scores, and their sign matches misclassification patterns (e.g., Moscow negative for `backend_general_dev`). P2 is supported as consistency between explanations and disparity metrics (Section 3.6). We do not treat IG as causal or legal proof. High- $|IG|$ tokens were pooled into the scrubbing and city-swap lexicon for Phase 3; scrubbing stacks then reach flip rates near zero (Table 4.2), while loss-only stacks stay near the baseline flip.

RQ3: How methods differ and what to deploy. RQ3 is answered by a trade-off map (Table 4.4). Across more than 25 tuned configurations (Phase 2) and combined runs (Phase 3), no checkpoint leads on utility, worst-case gap, and flip rate together, which confirms P4. P3 also holds: each family moves some axes and hurts others (GroupDRO lowers gap at a large F1 cost; scrubbing stacks lower flip without closing the TPR gap).

Phase 2–5 name axis-specific picks (numbers in Chapter 4): label smoothing for highest macro F1; R-Drop for the best regularizer gap with stable multi-seed behavior (Table 4.3); GroupDRO ($\eta = 0.1$) for lowest worst-case gap; Scrubbing + R-Drop for best F1 among low-flip models; Scrubbing + GroupDRO for minimum flip. Teams should state which axis matters most (Section 5.2) instead of looking for one unbiased classifier.

5.2 The Fairness–Utility–Robustness Triangle

The experiments organize competing objectives that HR and engineering teams often conflate:

- **Utility:** macro F1 (and accuracy) on the nine-class task;
- **Group fairness:** robust worst-case TPR gap across city groups;
- **Wording robustness:** city-swap flip rate on the evaluated subset (Section 3.5.5), a wording stress test.

Table 4.4 names one configuration per priority axis; Table 4.1 and Table 4.2 supply the underlying metrics. Three patterns matter for interpretation:

Gap and F1 trade off, but not linearly. Moving from the best regularizer gap among loss-based methods (R-Drop, 0.312) to the best GroupDRO gap (0.276) costs about six macro F1 points (Section 4.5). Smaller η or weighted sampling can soften that cost but do not remove the frontier.

Flip rate and TPR gap can diverge. Scrubbing-based models (Table 4.2) drive flip toward 0% while worst-case gaps stay near 0.32–0.36. Masking location tokens can stabilise predictions under our swap protocol without closing group TPR gaps. Teams that need stable decisions under city rewrites should still report TPR gaps.

Single-seed fairness rankings are fragile. Phase 4 (Table 4.3) shows label smoothing gap variance ± 0.044 across seeds, enough to change which method looks best for fairness. R-Drop has tighter F1 stability (± 0.005). Fairness claims should use multiple seeds or explain a single-seed choice.

Together, these patterns show that hiring models in practice need documented trade-offs across measurable axes, in line with P3 and P4.

5.3 Why Geographic Bias Dominates

The attribute breakdown in Phase 1 (Figure 4.1) motivates focusing debiasing on city groups. Three mechanisms, drawn from the data and error analysis, explain why gaps stay high (roughly 28–39% on tuned models; Table 4.1) even after extensive intervention:

Proxy vocabulary beyond city names. Resumes signal geography through employers, universities, projects, and regional phrasing, as well as city names. Scrubbing masks a fixed lexicon; GroupDRO and reweighting shift training loss but cannot remove every proxy.

Fragmented groups and finite test size. With 41 city groups, nine classes, and 5,510 test resumes, many group–class cells are sparse. The $\tau = 30$ support rule stabilises gaps but cannot add data where counts are low.

Labour-market structure. Roles cluster by region in historical data. That correlation can be informative for aggregate trends yet still produce unequal recognition across city groups when location should not drive the profession label. Our metrics report that tension; they do not settle the policy choice.

This explains why geography remains the binding attribute and why input masking and loss rebalancing alone hit a ceiling (Section 5.6).

5.4 Negative and Conditional Findings

Several outcomes contradict common practitioner shortcuts; they are reported in Chapter 4 and interpreted here.

Adversarial debiasing under high-cardinality geography. The exploratory gradient-reversal run in Section 4.2 (city group adversary, 41 groups, $\lambda = 0.5$) yields worst-case gap **42.9%** and macro F1 0.594, below the baseline on fairness and utility. With many city groups and thin per-group counts, the adversary may hurt the main task without hiding geography. **Recommendation:** avoid adversarial debiasing on high-cardinality location proxies without redesign.

Class-imbalance losses and group fairness. Focal loss, class-balanced CE, and training-time logit adjustment target class priors or hard examples, not city groups (Table 4.1). Class-balanced CE reaches gap 0.388; logit adjustment 0.371. Fixing class imbalance did not fix geographic TPR disparity on this dataset.

Post-hoc calibration limits. Post-hoc logit adjustment on the best label-smoothing checkpoint (Section 4.4.2) can raise F1 to 0.626 at $\tau = 0.3$ but worsens gap to 0.341–0.347; flip at $\tau = 0.5$ stays 7.0%, near the baseline. Score adjustment after training did not remove geographic bias in representations.

Combinations rarely beat single-method fronts. Primary stacks (Table 4.2) do not beat R-Drop on gap (0.312) or GroupDRO on gap (0.276); their value is mainly on the robustness axis (scrubbing) or as negative evidence for naive stacking.

These results delimit what the evaluated toolkit achieves without new data, new features, or architectural changes beyond the tested losses and scrubbing lexicon; methodological boundaries are summarised in Section 5.6.

5.5 Method Selection Guide

Table 5.1 translates Table 4.4 into deployment language: each row states a *binding constraint*, a recommended configuration, and the main cost on other axes. Practitioners should pick one row, document the trade-off for HR and legal stakeholders, and report metrics on all three axes where city-swap was run.

TABLE 5.1
Debiasing choice by organizational priority (metrics from Chapter 4; seed 42 unless noted)

Priority	Recommended configuration	Cost on other axes
Maximum macro F1	Label smoothing ($\epsilon=0.1$)	Worst-case gap $\sim 32\%$; city-swap flip $\sim 9\%$
Stable F1 and gap across seeds	R-Drop ($\alpha=1.0$); multi-seed report (Table 4.3)	Gap $\sim 31\%$; flip not the lowest
Lowest worst-case TPR gap	GroupDRO ($\eta=0.1$)	Macro F1 drops to ~ 0.56 ; flip not run for this checkpoint
Lowest city-swap flip	Scrubbing + GroupDRO ($\eta=0.1$) <i>or</i> Scrubbing + R-Drop	Gap stays $\sim 32\text{--}36\%$; F1 may fall vs. best single methods
Best F1 with very low flip	Scrubbing + R-Drop	Gap $\sim 32\%$; not the minimum TPR gap
Audit before debiasing	Baseline + Integrated Gradients (Phase 1)	No mitigation until bias path is documented
Do not use without redesign	Adversarial debiasing (41 city groups)	Gap can exceed baseline (42.9%)

Audit workflow. The intended workflow is: (1) measure utility and robust gaps on the full test set for every candidate checkpoint; (2) run city-swap only on family representatives and chosen stacks (Section 3.5.5); (3) retrain anchors on multiple seeds when fairness is a release criterion; (4) record the chosen row in Table 5.1 and the rejected alternatives. The appendix table (Table A.1) supports internal comparison across all 39 reported configurations without replacing the phased narrative in Chapter 4.

5.6 Scope and Limitations

Every study on real hiring data faces boundaries. Stating them clearly is part of an audit-first thesis: they tell the reader *where* the trade-off map applies and *what* would require new evidence. Metric definitions and protocols are in Chapter 3; here we summarise their implications for interpretation.

Data and external validity. The corpus is a public hh.ru coursework release [11], not an independent peer-reviewed benchmark (Section 3.2.1). It is concentrated in large cities, limited to nine IT supercategories, and tied to the metadata quality of that release. Our numbers describe this slice of the Russian IT labour market. Other countries, sectors, or label schemes need the same pipeline run on their own data before results transfer.

Fairness metrics. Robust TPR gaps with $\tau = 30$ are the main fairness statistic (Section 3.5.3). We do not report demographic parity, full equalized odds, calibration parity, or intersectional constraints. The work measures whether the correct class is assigned under a fixed argmax rule, group by group. That is actionable for model review; it is not a legal or moral verdict on “fair hiring” overall.

Geographic depth versus other attributes. Phase 1 measures city, gender, and age gaps for comparison (Figure 4.1). Phases 2–3 focus mitigation on geographic proxy bias because it is largest on the baseline. Gender and age inform the audit story but do not receive the same depth of debiasing search across 41 city groups.

Reproducibility of the search. Phase 2–3 tuning is led by seed 42 (Section 3.3.3). Multi-seed checks cover three anchors only (Table 4.3); for label

smoothing, worst-case gap varies by ± 0.044 across seeds, enough to change fairness rankings. Tables 4.1–4.4 reflect seed 42 unless a deployment team re-runs the search for release.

City-swap as a stress test. Flip rate comes from a controlled token-replacement protocol (Section 3.5.4), not from real relocation or full intersectional counterfactuals. Replacing city names can alter how plausible employers or education lines read. The triangle in Section 5.2 shows that low flip and low TPR gap are related but not the same goal.

Explainability. Integrated Gradients supported lexicon construction and qualitative checks against group gaps (Section 3.6). They are useful for audit; they are not stable causal evidence, and they can miss document-level or sub-word proxies.

Breadth of the method survey. More than 25 configurations were compared with one tuned representative per family, not an exhaustive multi-seed grid (Section 3.7). The gain is a structured map of trade-offs; finer hyperparameter search may move individual rankings.

Residual disparity and reporting scope. Flip-rate comparisons apply only to the city-swap subset (Section 3.5.5), not to every row in Table A.1. Even strong checkpoints leave gaps: GroupDRO reaches 0.276 worst-case TPR gap; Scrubbing + R-Drop reaches 0.09% flip with gap 0.324. Intersectional groups, per-group threshold rules, and broader post-hoc calibration were outside scope (Section 4.4.2).

Closing note. These limits narrow the claims, they do not erase them. Within the stated scope, the thesis delivers a repeatable five-phase audit, a documented trade-off map on realistic HR text, and explicit guidance when priorities conflict. Readers should treat the contribution at that level: evidence for informed deployment and further study, not proof that automated screening is unbiased in every sense.

5.7 Future Work

Natural extensions follow directly from the limits above: independent hiring corpora, intersectional gap metrics, wider seed studies, richer counterfactual tests, and IG-driven scrubbing at runtime. Representation learning and ensembles beyond the tested loss families may shift the Pareto frontier. In production, the priority is to embed the audit protocol in governance (periodic gap dashboards, multi-seed release gates, human review triggers) rather than to run a single debiasing sweep and treat the result as permanently fair.

Chapter 6

Conclusion

This work studied biased AI in recruitment: how to measure group disparity, how far standard debiasing moves utility, fairness, and wording robustness, and how to pick a deployment setting when those goals conflict. On 27,550 Russian resumes from HeadHunter (hh.ru), we ran the five-phase audit protocol (Chapter 3, Section 3.8) and evaluated more than 25 trained configurations plus selected combinations (Chapter 4; full metrics in Table A.1).

The main empirical finding is clear: **no tested checkpoint leads on utility, worst-case group fairness, and city-swap robustness together** (Table 4.4). Geographic bias is largest on the baseline; each debiasing choice improves one axis and costs another.

6.1 Completion of Research Tasks

The purpose of this work was addressed through the following task outcomes:

1. **Literature review.** We explored fairness definitions, debiasing families,

explainability tools, and recruitment-specific risks; working propositions P1–P4 (Section 1.7) translated this into testable expectations.

2. **Dataset and baseline.** We built a stratified train/validation/test split, a BERT-based nine-class classifier with inverse square-root reweighting by city group (Section 3.3), and reported baseline utility (macro F1 0.621) together with robust TPR gaps (Section 4.1.1).
3. **Bias measurement.** Phase 1 compared worst-case gaps across city, gender, and age (Figure 4.1: 32.9%, 12.4%, 18.2%), with geography as the binding attribute; Integrated Gradients gave diagnostic token evidence (Section 4.1.3), without causal proof.
4. **Debiasing implementation.** Phase 2 tuned representative checkpoints per method family (Table 4.1); Phase 3 trained primary and extended combinations; an exploratory adversarial run documented a high-cardinality failure mode (Section 4.2).
5. **Comparative evaluation.** All runs received accuracy, macro F1, and robust gaps on the full test set; city-swap flip rates were computed only for the subset in Section 3.5.5 (Table 4.2). Phase 4 reported multi-seed stability (Table 4.3) and post-hoc logit-adjustment limits (Section 4.4.2).
6. **Deployment guidance.** Phase 5 and Chapter 5 consolidated axis-specific optima (Table 4.4) and a priority-based selection table (Table 5.1).

6.2 Answers to Research Questions and Propositions

RQ1 (measurable bias). Yes. The baseline shows substantial group disparities under the robust metric ($\tau = 30$), with geography as the binding attribute (32.9% worst-case TPR gap vs. 18.2% age and 12.4% gender). Disparity persists under square-root city reweighting, consistent with bias from historical hiring patterns and proxy features in text [3]. **P1** is confirmed.

RQ2 (Integrated Gradients as audit evidence). IG attributions on baseline errors match the direction of group-level gaps; high- $|IG|$ tokens fed the scrubbing and city-swap lexicon in Phase 3. Explainability here supports diagnosis and targeted input edits before debiasing. **P2** is confirmed.

RQ3 (method differences and deployment fit). Methods sit in different parts of a three-axis map (utility, worst-case gap, flip rate). On seed 42 (Chapter 4): label smoothing for highest F1 (0.623); GroupDRO ($\eta = 0.1$) for lowest worst-case gap (0.276, F1 0.562); R-Drop for stable F1 and gap across seeds; Scrubbing + R-Drop for lowest flip (0.09%, F1 0.622) while the TPR gap stays high. No single model meets every priority; **P4** is confirmed. Families improve some metrics and hurt others; **P3** is confirmed.

6.3 Scientific Novelty Revisited

The contributions claimed in Section 1.5 are supported as follows:

1. **Five-phase audit protocol.** Phases 1–5 (diagnostic audit, debiasing benchmark, combinations and counterfactuals, validation, trade-off map) were executed on the HeadHunter-derived corpus with documented scope for city-swap (subset only) and reproducibility checks (multiple seeds on three anchors).
2. **Linking IG to group metrics and counterfactuals.** Token attributions motivated scrubbing; group gaps and flip rates were reported on the same checkpoints where city-swap was run, so disparity and wording sensitivity can be compared side by side.
3. **Explicit trade-off map.** Tables 4.4 and 5.1 list axis-specific optima and organizational priorities without merging fairness into one score.
4. **Documented failure modes.** Adversarial debiasing worsened worst-case gap to 42.9% with 41 city groups; class-imbalance and post-hoc calibration did not fix geographic TPR disparity; single-seed fairness rankings were unstable for label smoothing (Table 4.3). These boundaries are analyzed in Section 5.4.

6.4 Practical Implications

For HR and compliance teams, three habits follow from the results: measure the baseline and run IG before debiasing; state which constraint matters most (accuracy, worst-case gap, or stable predictions under city rewrites); and report several metrics and seeds when fairness gates release. Table 5.1 (Chapter 5) lists concrete picks: R-Drop ($\alpha = 1.0$) for reproducible utility with moderate gap change; Scrubbing + R-Drop when geographic rewrite stability is required; GroupDRO ($\eta = 0.1$)

when gap reduction outweighs F1.

6.5 Limitations and Outlook

This paper does not claim a universally fair resume classifier. It claims something more modest and, for practice, more useful: on a large HeadHunter-derived corpus, bias is measurable, geographic proxy effects dominate, and more than 25 debiasing setups move utility, worst-case TPR gap, and city-swap flip rate in predictable tension. Those conclusions rest on explicit scope choices: a coursework corpus rather than an external benchmark; robust TPR gaps as the primary fairness statistic; deep mitigation work on city-related disparity with gender and age mainly in Phase 1; seed 42-led tuning checked on three multi-seed anchors; city-swap as a controlled wording test; Integrated Gradients as audit support rather than causal proof; and a broad but shallow survey of method families (Section 5.6).

Within these limits, the contribution is practical. Teams can run a repeatable five-phase audit, see where methods fail as well as succeed, and choose a checkpoint based on what matters most: accuracy, worst-case TPR gap, or stable predictions when city names in a CV are rewritten. Later work can add new datasets, intersectional metrics, more random seeds, and stronger counterfactual tests. In production, the same audit should become part of HR governance: fairness is measured on a schedule, not assumed once at model release.

Appendix A

Implementation Reference

This appendix gives a concise pseudocode view of the five-phase audit protocol (Section 3.8) and a full results catalog (Table A.1). Classification metrics are computed on the test set for every checkpoint; city-swap flip is run only on the subset defined in Section 3.5.5.

A.1 Five-phase audit protocol

```
ALGORITHM ResumeScreeningAudit

CONSTANTS
    K <- 9                // profession classes
    MAX_LEN <- 128        // BERT tokens
    BATCH <- 8, LR <- 2e-5, EPOCHS <- 2
    MIN_SUPPORT <- 30     // robust TPR gap (tau)
    IG_STEPS <- 25
    LEXICON                // city/age tokens from Phase-1 IG
    SWAP_CITIES            // fixed list for counterfactual test

DATA
    splits: train / val / test with (text, label, city, gender, age)
```

```

CONFIG: {recipe, hparams, scrub_train?, scrub_infer?}
SEED

```

```

-----
FUNCTION SqrtCityWeights(train):

```

```

    FOR each city c: w[c] <- 1 / sqrt(count_c)
    normalize w; assign w[i] <- w[city_i]
    RETURN sample weights

```

```

FUNCTION Scrub(text):

```

```

    FOR token in LEXICON ordered by length (longest first):
        text <- replace(text, token, "[MASK]", ignore_case)
    RETURN text

```

```

FUNCTION Encode(text, CONFIG):

```

```

    IF CONFIG.scrub_train: text <- Scrub(text)
    RETURN Tokenize(text, max_len=MAX_LEN)

```

```

FUNCTION SwapCity(text, target):

```

```

    RETURN replace city tokens in text with target (case preserved)

```

```

-----
PROCEDURE Train(CONFIG, SEED):

```

```

    set_seed(SEED)
    model <- BERT_for_sequence_classification(K)
    weights <- SqrtCityWeights(train)    // except weighted-sampler recipe

    texts <- train.text with optional Scrub per CONFIG
    trainer <- TrainerFor(CONFIG.recipe, weights, CONFIG.hparams)
    trainer.fit(model, texts, val, lr=LR, batch=BATCH, epochs=EPOCHS,
                select_checkpoint = min validation loss)
    RETURN model

```

```

-----
FUNCTION Predict(model, texts, CONFIG):

```

```

    IF CONFIG.scrub_infer AND NOT CONFIG.scrub_train:
        texts <- map Scrub over texts

```

```

RETURN argmax model(logits(texts))

FUNCTION RobustGaps(y_true, y_pred, groups):
  TPR[g,c] <- one-vs-rest true positive rate per group g, class c
  FOR each class c:
    G_c <- {g : support(g,c) >= MIN_SUPPORT}
    gap[c] <- max_{g in G_c} TPR[g,c] - min_{g in G_c} TPR[g,c]
  RETURN max_c gap[c], mean_c gap[c]

FUNCTION CitySwapFlip(model, test, CONFIG):
  y0 <- Predict(model, test.text, CONFIG)
  flips <- empty list
  FOR target in SWAP_CITIES:
    t_sw <- [SwapCity(t, target) for t in test.text]
    IF CONFIG.scrub_infer: t_sw <- map Scrub over t_sw
    y1 <- Predict(model, t_sw, CONFIG)
    append flips with indicator(y0 != y1) per sample
  RETURN mean(flips) // fraction of (sample, target) pairs that flip

FUNCTION IntegratedGradients(model, text, class_c):
  attributions <- LayerIG on embedding layer, n_steps=IG_STEPS,
    baseline = zero embeddings, target = class_c
  RETURN token-level attribution scores

-----
MAIN // five phases; default SEED = 42 for search

// --- Phase 1: diagnostic audit ---
baseline <- Train(baseline_CONFIG, SEED=42)
(acc, f1, worst, macro) <- metrics on test + RobustGaps(...)
LEXICON <- high-|IG| city/age tokens from baseline errors
// city-swap on baseline only (legacy checkpoint: flip ~ 7.7%)

// --- Phase 2: debiasing benchmark (>25 hyperparameter runs) ---
FOR CONFIG in single_method_grid:
  model <- Train(CONFIG, SEED=42)
  record acc, f1, worst_gap, macro_gap on full test set

```

```
// no city-swap here (too many trials)
END FOR
pick best_per_family representatives for later flip test

// --- Phase 3: combinations ---
FOR CONFIG in {scrub+LS, LS+R-Drop, scrub+R-Drop, CBCE+LS, scrub+GroupDRO,
...}:
    model <- Train(CONFIG, SEED=42)
    record classification metrics on test
END FOR
FOR checkpoint in city_swap_subset:    // Section subsec:city_swap_scope
    record CitySwapFlip(checkpoint)
END FOR

// --- Phase 4: validation ---
FOR anchor in {R-Drop, Label Smoothing, Weighted Sampler}:
    FOR seed in {7, 13, 21}:
        model <- Train(anchor, seed)
        record test metrics
    aggregate mean +/- std over seeds
END FOR

// --- Phase 5: trade-off map ---
FOR tau in {0.3, 0.5, 0.7}:            // post-hoc, no retrain
    adjust saved logits of best label-smoothing checkpoint
    record acc, f1, gaps; city-swap at tau=0.5 only
END FOR
RETURN Pareto-style recommendations by priority (utility / gap / flip)

END ALGORITHM
```

A.2 Full model catalog

Table A.1 lists 39 configurations: the baseline, 29 tuned challenger checkpoints from the Phase 2 grid, six Phase 3 combination models, and three post-hoc logit-adjustment settings on the best label-smoothing run. Rows are sorted by worst-case TPR gap (ascending), then macro F1 (descending). Regenerate this table from project CSVs with:

```
python3 scripts/gen_appendix_models_table.py
```

TABLE A.1
**Full catalog of trained checkpoints (test set, robust TPR gap with $\tau=30$).
 City-swap flip (%) only for the subset in Section 3.5.5; “—” = not run**

Configuration	Acc.	F1	Gap	M. gap	Flip (%)
Baseline (BERT + $\sqrt{n}$ city reweight)	0.609	0.621	0.329	0.116	7.70
GroupDRO ($\eta=0.1$, 2 ep)	0.546	0.562	0.276	0.131	—
GroupDRO ($\eta=0.05$, 2 ep)	0.575	0.590	0.288	0.122	8.75
Weighted city sampler ($p=1.5$, 2 ep)	0.561	0.561	0.288	0.107	—
Focal loss ($\gamma=20$, 3 ep)	0.606	0.615	0.300	0.146	—
R-Drop ($\alpha=1.0$, 2 ep)	0.607	0.620	0.312	0.127	6.21
Focal loss ($\gamma=15$, 3 ep)	0.606	0.613	0.312	0.139	—
Label smoothing + GroupDRO ($\epsilon=0.1$, $\eta=0.1$, 3 ep)	0.574	0.587	0.312	0.119	—
Label smoothing ($\epsilon=0.1$, 2 ep)	0.608	0.623	0.318	0.136	8.91
Label smoothing ($\epsilon=0.05$, 2 ep)	0.605	0.620	0.324	0.128	—
Focal loss ($\gamma=1$, 2 ep)	0.604	0.619	0.324	0.134	8.68
Scrub + R-Drop ($\alpha=1.0$, 2 ep)	0.606	0.622	0.324	0.130	0.09
Scrub + label smoothing ($\epsilon=0.1$, 2 ep)	0.605	0.620	0.324	0.127	0.16

continued on next page

TABLE A.1 (continued)

Configuration	Acc.	F1	Gap	M. gap	Flip (%)
Scrub + GroupDRO ($\eta=0.1$, 2 ep)	0.577	0.588	0.324	0.127	0.00
R-Drop ($\alpha=0.5$, 2 ep)	0.606	0.618	0.329	0.116	—
Weighted city sampler ($p=1.0$, 2 ep)	0.578	0.583	0.329	0.131	—
R-Drop ($\alpha=2.0$, 2 ep)	0.604	0.621	0.335	0.138	—
Class-balanced CE ($\beta=0.999$, 2 ep)	0.601	0.615	0.335	0.136	—
Label smoothing + GroupDRO ($\epsilon=0.05$, $\eta=0.05$, 3 ep)	0.588	0.600	0.335	0.132	—
Label smoothing + GroupDRO ($\epsilon=0.1$, $\eta=0.05$, 3 ep)	0.586	0.594	0.335	0.127	—
Label smoothing + R-Drop ($\epsilon=0.1$, $\alpha=1.0$, 2 ep)	0.607	0.620	0.335	0.130	6.04
Post-hoc logit adj. ($\tau = 0.3$) on best label smoothing	0.610	0.626	0.341	0.130	—
CBCE + label smoothing ($\epsilon=0.1$, $\beta=0.99$, 2 ep)	0.603	0.617	0.341	0.124	8.62
Label smoothing ($\epsilon=0.15$, 2 ep)	0.599	0.616	0.341	0.141	—
Label smoothing ($\epsilon=0.1$, 3 ep)	0.606	0.615	0.341	0.119	—
Focal loss ($\gamma=3$, 2 ep)	0.600	0.617	0.347	0.143	—
GroupDRO ($\eta=0.2$, 2 ep)	0.521	0.533	0.347	0.125	—
Post-hoc logit adj. ($\tau = 0.7$) on best label smoothing	0.606	0.624	0.347	0.125	—
Post-hoc logit adj. ($\tau = 0.5$) on best label smoothing	0.606	0.624	0.347	0.125	7.02
Label smoothing ($\epsilon=0.05$, 3 ep)	0.601	0.608	0.353	0.152	—
Weighted city sampler ($p=0.5$, 2 ep)	0.592	0.602	0.353	0.148	—
Logit adjustment (train) ($\tau=1.5$, 2 ep)	0.570	0.596	0.353	0.121	—
Logit adjustment (train) ($\tau=1.0$, 2 ep)	0.595	0.610	0.359	0.142	—
Scrub + GroupDRO ($\eta=0.05$, 2 ep)	0.588	0.601	0.359	0.117	0.15

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TABLE A.1 (continued)

Configuration	Acc.	F1	Gap	M. gap	Flip (%)
Focal loss ($\gamma=10$, 3 ep)	0.605	0.613	0.365	0.128	—
Class-balanced focal ($\gamma=20$, $\beta=0.999$, 2 ep)	0.603	0.612	0.365	0.139	—
Logit adjustment (train) ($\tau=0.5$, 2 ep)	0.599	0.617	0.371	0.126	—
Focal loss ($\gamma=2$, 2 ep)	0.596	0.614	0.371	0.134	—
Class-balanced CE ($\beta=0.99$, 2 ep)	0.607	0.621	0.388	0.137	5.77

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